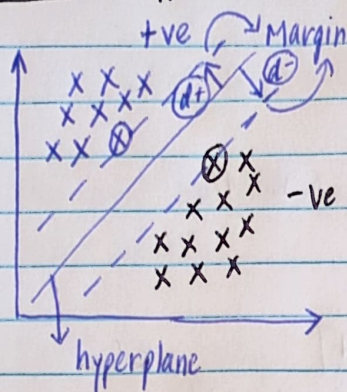
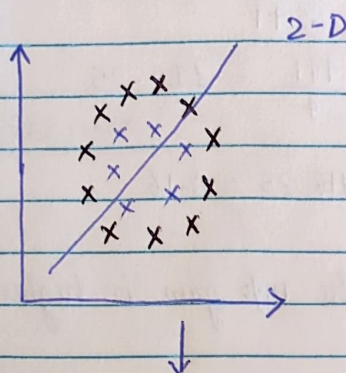


SVM (Support Vector Machine)



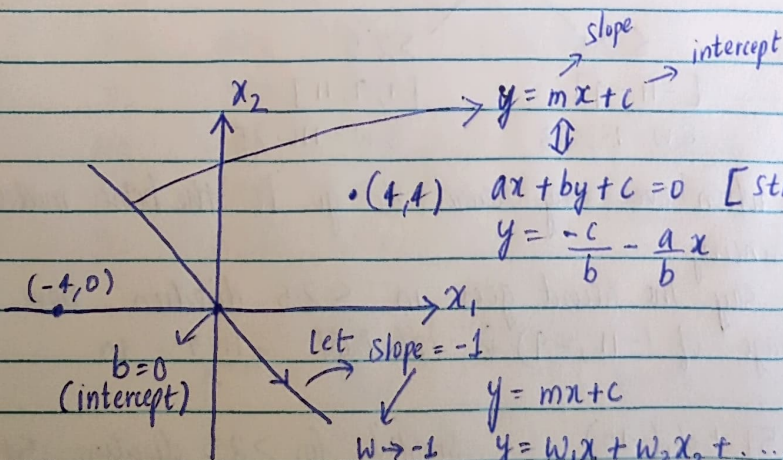
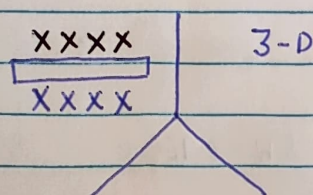
• we select the hyperplane which has maximum marginal distance

• $\otimes$ and $\otimes$ are called as support vectors



• SVM kernels try to convert Low dimensional into High dimensional

2-D to 3-D



• (4, 4) $ax + by + c = 0$ [straight line eqn]

$$y = \frac{-c}{b} - \frac{a}{b}x$$

$$m = -a/b$$

$$c = -c/b$$

$$y = mx + c$$

$$y = w_1x_1 + w_2x_2 + \dots + b$$

$$y = w^T x + b$$

$$y = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \begin{bmatrix} -4, 0 \end{bmatrix}$$

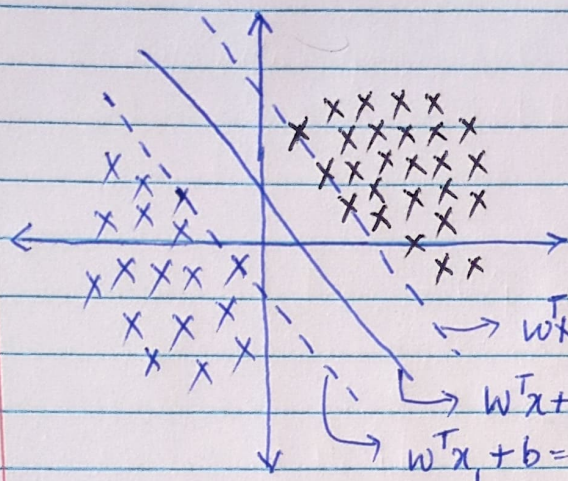
$$= 4 \text{ (+ve value)}$$

Any point below the line has +ve value of y.

$$y = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \begin{bmatrix} 4, 4 \end{bmatrix}$$

$$= -4 + 0 = -4 \text{ (-ve value)}$$

Similarly, for any point above the line has -ve y value.



$$ax + by + c = 0$$

$$\uparrow$$

$$w^T x + b = 0$$

$$w^T x_1 + b = 1$$

$$(-) w^T x_2 + b = -1$$

$$w^T (x_1 - x_2) = 2$$

$$\frac{w^T (x_2 - x_1)}{\|w\|} = \frac{2}{\|w\|}$$

We need to maximise $\frac{2}{\|w\|}$

$(\tilde{w}, \tilde{b})$ max $\frac{2}{\|w\|} \rightarrow$ optimization function
 update $\downarrow$
 $\|w\| \rightarrow$ to increase marginal plane

such that $y_i \begin{cases} +1 & w^T x + b \geq 1 \\ -1 & w^T x + b \leq -1 \end{cases}$

Major aim $y_i (w^T x_i + b) \geq 1$

Maximise $\frac{2}{\ w\ }$	$\Rightarrow$	Minimise $\frac{\ w\ }{2}$
$(\tilde{w}, \tilde{b})$		$(\tilde{w}, \tilde{b})$

$$\min_{(\tilde{w}, \tilde{b})} \frac{\|w\|}{2} + c_i \sum_{i=1}^n \xi_i$$

how many errors we can have

value of the error

summation of the distance of the wrong datapoints

